

11. SYNTH TRACK PARAMETERS

Here follows a description of all of the parameters that are available on the synth tracks **PARAMETER** pages. The parameters on the **TRIG** page are saved with the pattern, and not with the sound. You can lock the synth track parameters to other settings on any step of the pattern by first pressing and holding a **[TRIG]** key, then changing the parameters' settings with the **DATA ENTRY** knobs. For more information, please see "10.9.1 **PARAMETER LOCKS**" on page 40.

11.1 EDITING THE SYNTH TRACK PARAMETERS

There are six **PARAMETER** page groups for the synth tracks. Press the **[PARAMETER]** keys to access the synth tracks **PARAMETER** pages. The **SYN2**, **FLTR**, **AMP** and **LFO** parameter groups has two pages. To access page one, press the **[PARAMETER]** key once. To access page two, press the **[PARAMETER]** key twice. Use the **DATA ENTRY** knobs **A-H** to change the parameters. Press and hold a **[PARAMETER]** key to see all the parameter settings on the **PARAMETER** page.



- You can also randomize the parameter settings on a specific **PARAMETER** page on a Synth track. Press **[PARAMETER]** key + **[YES]** to randomize all parameters on that page. Every time you press this key combination, the parameters randomize in a new way.
- Press **[PARAMETER]** key + **[NO]** to reset the **PARAMETER** page to its last saved state.
- Press and hold a **[PARAMETER]** key to see all the parameter settings on the **PARAMETER** page.

11.2 TRIG PARAMETERS PAGE

Here you find the parameters for the note trigs. Use the **DATA ENTRY** knobs to change the settings. The settings on the **TRIG PARAMETERS** page is not saved with the Sound. They are saved with the pattern.

Press **[TRIG PARAMETERS]** to access this parameter page.



11.2.1 ROOT

Root sets the default note value that the note trigs placed in **GRID RECORDING** mode have. If you change the root note and have additional notes added on the same trig, then the additional notes are offset and transposed accordingly. For more information, please see "10.2.3 **GRID RECORDING USING TRACK NOTE METHOD**" on page 33. (C0–G10)

11.2.2 VEL

Trig Velocity controls the velocity of the notes the MIDI track sends out. A setting of 0 equals a **NOTE OFF** command. (1–127)

11.2.3 LEN

Trig Length sets the duration of the notes. When a note has finished playing a **NOTE OFF** command is sent. The **INF** setting equals infinite note length. (0.125–128, **INF**)

11.2.4 PROB

TRIG PROBABILITY sets the probability that the trigs on the track plays or not. The probability outcome is re-evaluated every time a trig is set to play. The default setting is 100%, meaning that all the trigs on the track plays every time. (0%–100%)

This parameter changes temporarily to display and control **COND** (Trig Condition) when you add a conditional lock. For more information, please see "10.9.3 **TRIG CONDITIONS AND CONDITIONAL LOCKS**" on page 40.



If you have placed a conditional lock on a trig in the sequencer, the trig condition overrides the Trig Probability setting. For example, if you set PROB to 70% and a FILL trig condition on a trig, that trig will only (and always) play when the Digitone is in FILL mode. For more information, please see “10.9.3 TRIG CONDITIONS AND CONDITIONAL LOCKS” on page 40.

11.2.5 FLT.T

Filter Trig controls if the filter envelope is triggered or not. (ON, OFF)

11.2.6 LFO.T

LFO Trig controls if the LFO is triggered or not. (ON, OFF)

11.2.7 PTIM

Portamento Time sets the time for the portamento (1–127)

11.2.8 PORT

Portamento turns the portamento on/off (ON, OFF)

For more information, please see “9.5.5 PORTAMENTO” on page 29.

11.3 SYN1 PAGE 1

The parameters on the SYN1 page 1 control various aspects of the FM synthesis. You can find more information about the Digitone FM synthesis in “APPENDIX A: THE DIGITONE FM SYNTHESIS” on page 90 and “A.7 SYN1 PAGE 1 PARAMETERS OVERVIEW” on page 94.

Press [SYN1] once to access this parameter page.



11.3.1 ALGO

Algorithm selects the set structure of how the four operators are connected to each other. (1–8) For more information, please see “A.3 ALGORITHMS” on page 91.

11.3.2 RATIO C

Sets the frequency ratio for operator C. (0.25–16.0) For more information, please see “A.4 FM RATIOS” on page 91.

11.3.3 RATIO A

Sets the frequency ratio for operator A. (0.25–16.0) For more information, please see “A.4 FM RATIOS” on page 91.

11.3.4 RATIO B

Sets the frequency ratio for operator B1 and B2. The minimum value for B1 and B2 is .25. As you turn the encoder, B2 increases until it reaches the max (16). It then starts over from .25 and B1 increases to the next value (0.5). This revolving behavior continues until both operators reach the maximum value. This parameter behavior is similar to the movement of the hands on a watch. (0.25–16.0) For more information, please see “A.4 FM RATIOS” on page 91.

11.3.5 HARM

Harmonics controls waveform of the operators, C, A, and B1. The parameter is bipolar. Negative parameter values change the harmonics of operator C, while positive parameter values change the harmonics of operators A and B1. (-26.00–26.00) For more information, please see “A.6 HARMONICS” on page 93.

11.3.6 DTUN

Detune offsets the ratio of operator A and B2. Up until a parameter value of around 64, the offset is very slight to achieve subtle movement and drifting. Above 64 the operators start to detune more heavily. (0.00–127.00)

11.3.7 FDBK

Feedback sets the amount of self modulation of the operator that has feedback. This operator is shown in the algorithm on the screen with a feedback loop in its upper left corner. (0.00–120.00). For more information, please see “A.2 THE OPERATORS” on page 90.

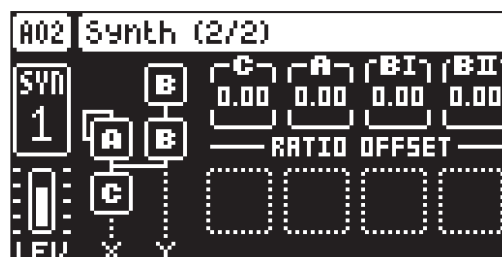
11.3.8 MIX

Each algorithm has two carrier outputs (X and Y) that come from two different operators depending on what algorithm you chose. Use the **MIX** parameter to mix between these outputs so that you can cross-fade between two separate timbres. (-64–63). For more information, please see “A.3 ALGORITHMS” on page 91.

11.4 SYN1 PAGE 2

The parameters on the SYN1 page 2 controls the operators ratio offsets.

Press [SYN1] twice to access this parameter page.



11.4.1 RATIO OFFSET C, A, B1, B2

Adds offsets to the ratios of each of the four operators.

11.5 SYN 2 PAGE 1

The parameters on the SYN2 pages control various aspects of the FM synthesis, mainly the amount of frequency modulation together with the operator envelopes and their behavior.

The FM engine has two operator envelopes. One is for operator group A, and one is for group B (B1 and B2). The envelopes are essentially expanded AD (Attack Decay) envelopes, but with an added adjustable end level (the amplitude level the sound reaches at the end of the decay stage). For more information, please see “A.5 THE OPERATOR ENVELOPES” on page 92.

Press [SYN2] once to access this parameter page.



11.5.1 ATK A

Attack Time sets the length of the attack phase of the modulation envelope for operator A. (0–127)

11.5.2 DEC A

Decay Time sets the length of the decay phase of the modulation envelope for operator A. (0–127)

11.5.3 END A

End Level sets the end level of the modulation envelope for operator A. (0–127)

11.5.4 LEV A

Level sets the modulation amount from operator A. (0–127)

11.5.5 ATK B

Attack Time sets the length of the attack phase of the modulation envelope for operator group B (B1 and B2). (0–127)

11.5.6 DEC B

Decay Time sets the length of the decay phase of the modulation envelope for operator group B (B1 and B2). (0–127)

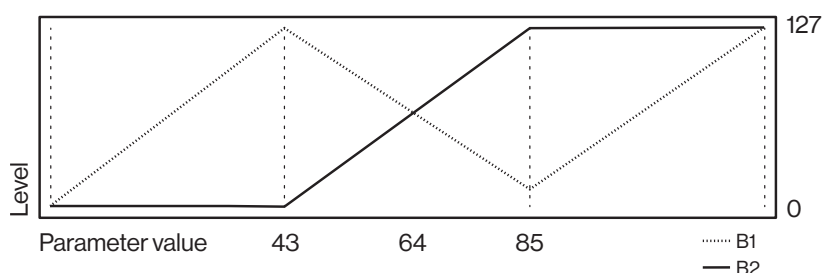
11.5.7 END B

End Level sets the end level of the modulation envelope for operator group B (B1 and B2). (0–127)

11.5.8 LEV B

Level sets the modulation amount from operator group B (B1 and B2). (0–127)

The **LEVEL** parameter for B is macro mapped to both operator B1 and B2 and control their modulation amount as per this graph:



If you wish to use frequency modulation, it is important that you turn the LEV parameters up, since they set the amount of frequency modulation in the FM engine.

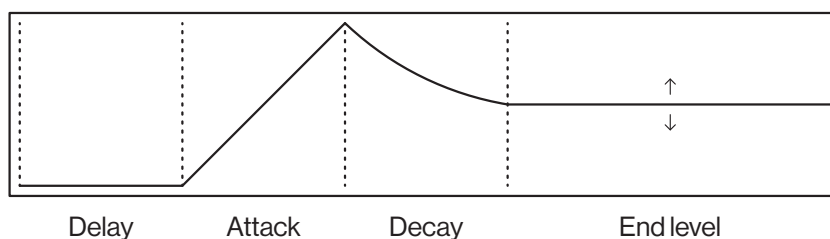
11.6 SYN2 PAGE 2

The parameters on the SYN2 pages control various aspects of the FM synthesis, mainly the amount of frequency modulation together with the operator envelopes and their behavior.

Press [SYN2] twice to access this parameter page.

**11.6.1 ADEL**

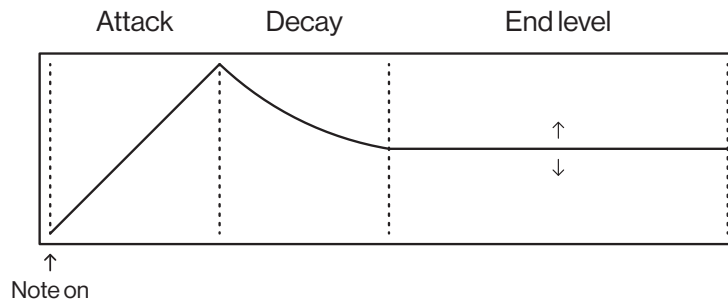
Envelope delay sets the time before attack phase of the modulation envelope for operator A starts. (0–127)

**11.6.2 ATRG**

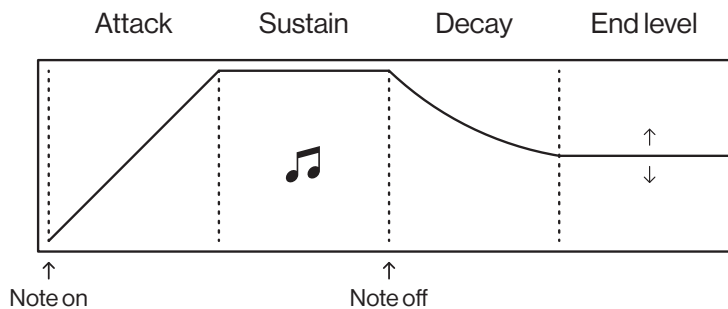
Envelope trig sets the trig behavior of the operator envelopes. The envelopes can be either triggered or gated - making it either an ADE (Attack Decay End) or an ASDE (Attack Sustain Decay End) envelope.

The sustain phase does not have an adjustable envelope level. It is instead the **LEV** parameter that sets the sustain level. The note length defines the length of the sustain phase.

Triggered (ATRG/BTRG ON)



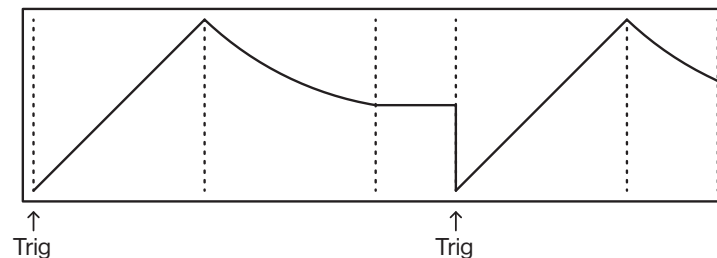
Gated (ATRG/BTRG OFF)



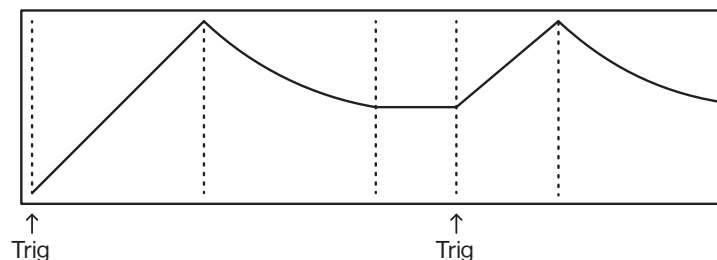
11.6.3 ARST

Envelope reset sets if the envelopes should reset or not when they are retriggered.

Reset on (ARST/BRST ON)



Reset off (ARST/BRST OFF)



11.6.4 PHRT

Phase reset sets if the operators phase are reset to start at 0 or not when they are triggered.

OFF do not reset any operators

ALL resets all operators

C resets operator C

A+B resets operator A, B1, and B2

A+B2 resets operator A and B2

11.6.5 BDEL

Same as **ADEL** but for operator group B (B1 and B2).

11.6.6 BTRG

Same as **ATRG** but for operator group B (B1 and B2).

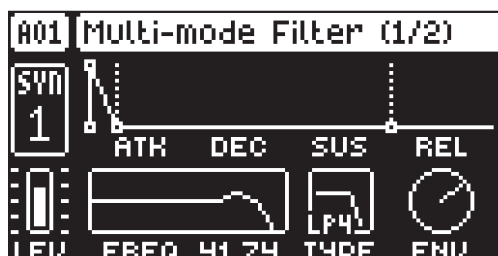
11.6.7 BRST

Same as **ARST** but for operator group B (B1 and B2).

11.7 FLTR PAGE 1

On **FILTER** page 1, you find all parameters that control the Multimode filter and its associated envelope. The Multimode filter and the Base-Width filter are placed in series with the Base-Width filter before the Multimode filter.

Press **[FLTR]** once to access this parameter page.



11.7.1 ATK

Attack Time sets the length of the attack phase of the filter envelope. (0-127)

11.7.2 DEC

Decay Time sets the length of the decay phase of the filter envelope. (0-127)

11.7.3 SUS

Sustain Level sets the sustain level of the filter envelope. (0-127)

11.7.4 REL

Release Time sets the length of the release phase of the filter envelope. (0-126, INF)

11.7.5 FREQ

Frequency sets the cutoff frequency of the multimode filter. (0.00-127.00)

11.7.6 RESO

Resonance sets the resonance behavior of the filter. Resonance introduces a peak in the spectrum at the cutoff frequency. (0.00-127.00)

11.7.7 TYPE

Filter Type selects the type of filter. (OFF, 2-pole (12 dB) Lowpass, 2-pole (12 dB) Highpass, 4 pole (24 dB) Lowpass)

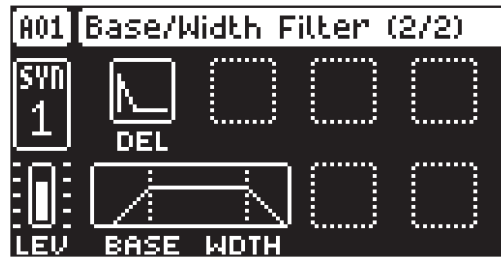
11.7.8 ENV

Env. Depth sets the amount of cutoff frequency modulation from the filter envelope. The parameter is bipolar, both negative and positive modulation depths are available. Range (-64.00-63.00)

11.8 FLTR PAGE 2

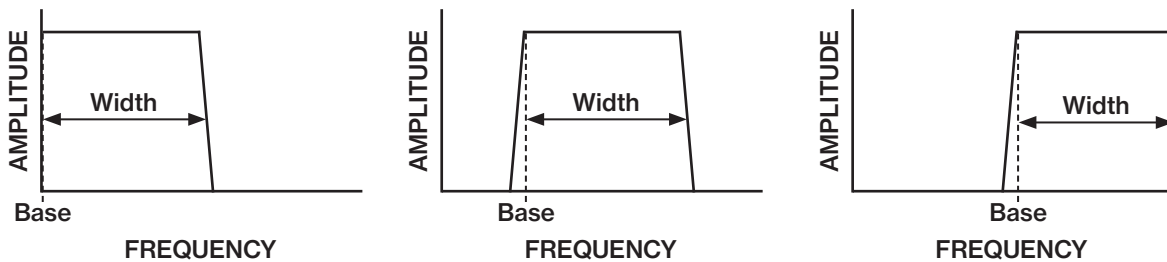
On **FILTER** page 2, you find the parameters that control the 1-pole (6 dB) Base-Width filter. The Base-Width filter and the Multimode filter are placed in series with the Base-Width filter before the Multimode filter.

Press **[FLTR]** twice to access this parameter page.



The base-width filter is basically a Highpass filter and a Lowpass filter connected in series. The filters **BASE** and **WIDTH** parameters define the base-width filters frequency range.

Examples of how the **BASE** and **WIDTH** parameters affect the filter frequency range:



- With **BASE** set to 0, the filter functions as a low pass filter with **WIDTH** adjusting the frequency range.
- With **WIDTH** set to 127, the filter functions as a high pass filter with **BASE** adjusting the frequency range.
- With **BASE** set to 0 and **WIDTH** set to 127 the filter does not affect the sound.

11.8.1 DEL

Envelope delay sets the time before the attack phase of the filter envelope starts. This parameter is part of the filter envelope that controls the multimode filter. You can also press and hold **[FUNC]** and then turn **DATA ENTRY** knob A to quickly access this parameter from FILTER page 1.

11.8.2 BASE

Sets the base frequency of the filter. (0–127)

11.8.3 WIDTH

Sets the frequency width above the base frequency. (0–127)

11.9 AMP PAGE 1

The AMP page 1 controls parameters for the amplitude envelope, overdrive, tone, panning and volume.

Press **[AMP]** once to access this parameter page.



11.9.1 ATK

Attack Time sets the length of the attack phase of the amp envelope. (0–127)

11.9.2 DEC

Decay Time sets the length of the decay phase of the amp envelope. (0–127)

11.9.3 SUS

Sustain Level sets the sustain level of the amp envelope. (0–127)

11.9.4 REL

Release Time sets the length of the release phase of the amp envelope. (0–126, INF)

11.9.5 DRV

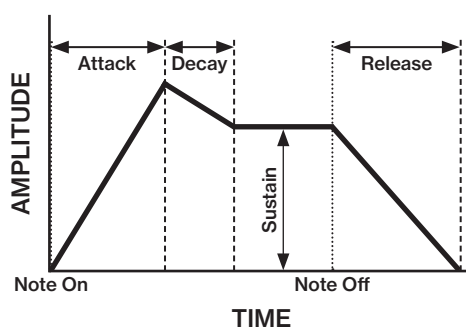
Overdrive sets the amount of overdrive and clipping distortion of the signal entering the filter. (0.00–127.00)

11.9.6 PAN

Pan positions the sound in the stereo field. The knob function is bipolar with 0 being the balanced mid-point, L64 sends all sound to the left channel, and R63 sends all sound to the right channel. (L64–R63)

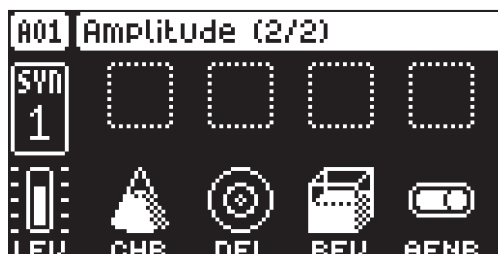
11.9.7 VOL

Volume sets the amount volume of the sound. This parameter is independent of the overall track level. (0.00–127.00)

**11.10 AMP PAGE 2**

The AMP page 2 controls parameters for the effects sends and amp envelope reset volume.

Press [AMP] twice to access this parameter page.

**11.10.1 CHR**

Chorus Send sets the amount of sound sent through to the Chorus effect. For more information, please see “13. FX PARAMETERS” on page 61. (OFF, 0.01–127.00)

11.10.2 DEL

Delay Send sets the amount of sound sent through to the Delay effect. For more information, please see “13. FX PARAMETERS” on page 61. (OFF, 0.01–127.00)

11.10.3 REV

Reverb Send sets the amount of sound sent through to the Reverb effect. For more information, please see “13. FX PARAMETERS” on page 61. (OFF, 0.01–127.00)

11.10.4 AENR

Amp envelope reset sets the amp envelope behavior:

ON resets the envelopes for each consecutive trig (default).

OFF does not reset the envelopes for each consecutive trig. It instead lets the envelope cycle complete.